

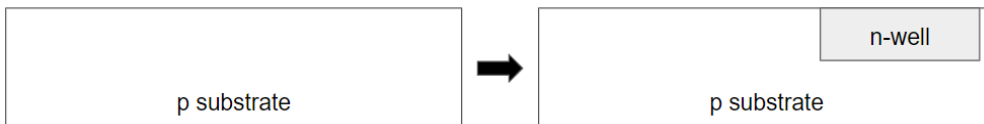
Semester: **Spring 2022**
Course Code: **CSE460**
Course Title: **VLSI Design**

Midterm Exam
Full Marks: **15 x 3 = 45**
Time: **1 hour 40 minutes**
Date: **11th March 2022**

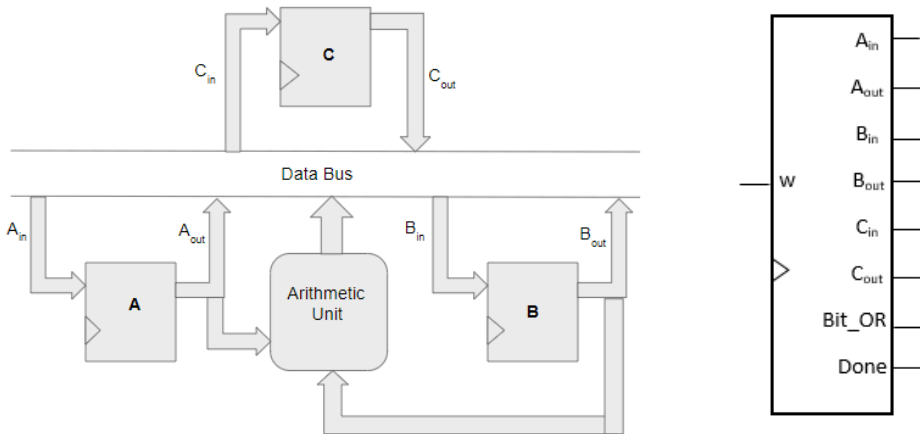
Student ID:	Name:	Section:
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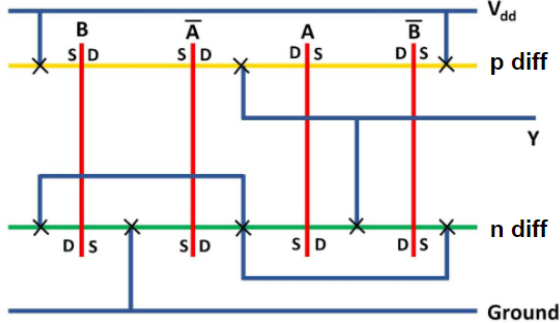
[Answer any **THREE** questions out of **FOUR**. Each question carries equal marks.]

[After the exam, the question paper should be turned in along with the answer script.]

1. (CO1)	To fabricate a CMOS inverter, first, we start with a blank wafer (figure 1) and then deposit an n-well <i>inside</i> that blank wafer (figure 2). The cross-sectional view of the process is shown below:  F F i i g g u u r r e e 1 2 : : c c r r o o s s s s - - s s e e c c t t i i o o
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	n n v v i i e e w w o o f f t t h h e e i w n a i f t e i r a a l f b t l e a r n s k u w c a c f e e s r s f u l l y d e p o s i t i n g t h e n	
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	- w e l l Note: you may assume any arbitrary layout that is consistent with figure 2.		
	(a)	Draw an n-well mask that can be used to produce figure 2.	3
	(b)	Briefly describe the fabrication steps involved in this process of depositing an n-well inside a blank wafer.	12
2. (CO2)	(a)	Design a 2 to 1 multiplexer that will pass either input A or B to the output F depending on the logic state of the control pin S . Implement your design using basic logic gates (AND, OR, INVERT) and determine the total number of nMOS and pMOS transistors required to implement the multiplexer	7
	(b)	Design the same multiplexer described in (a) using transmission gates only and calculate the total number of nMOS and pMOS transistors required to implement your design. Is this design restoring or non-restoring? Why?	8
3. (CO2)	 Here, a digital system comprising 3 registers (A, B, C) and an arithmetic unit is shown (left), along with the system's control circuit (right).		

	The arithmetic unit performs bitwise OR operation between the contents of registers A and B if the control signal Bit_OR = 1. The result of the bitwise OR operation is readily available in the data bus. For this problem, you have to design a Moore-type control circuit which performs bitwise OR operation between the contents of registers A and C; saves the result in register A. The process is initiated when the input w = 1. The operation should be completed once initiated, regardless of the value of w. Once completed, the “Done” signal should be set high, or Done = 1. If the “in” pin of a register (A_{in}, B_{in}, or C_{in}) is high, then the corresponding register can save the data available in the data bus. Similarly, any register’s saved data can be made available to the data bus by setting its “out” pin (A_{out}, B_{out}, or C_{out}) high. <i>[Hint: to accomplish the task, first you need to <u>transfer</u> the contents of register C to B, then <u>activate</u> the arithmetic unit by setting Bit_OR to high, and finally <u>transfer</u> the result to register A and set Done to high.]</i>	
(a)	Design a state diagram for the control circuit. (Reset and transitions should be clearly marked.)	7
(b)	Derive a state assigned table from (a).	4
(c)	Implement the FSM using D flip-flops.	4
4. (CO2)	(a) Sketch a transistor-level schematic and the corresponding stick diagram for a compound CMOS gate that implements the function: $F = \overline{(A + B + C)} \cdot \overline{D}$. Estimate the area of the logic gate from the stick diagram in terms of λ .	10
	(b) From the stick diagram given below, draw the transistor level schematic of the equivalent CMOS logic gate and determine the function “Y”. 	5